

Budgeting

Managing Money

2026

Preliminary Mathematics Standard

Outline

Ongoing Costs of a Vehicle

Household Bills

Personal Budgeting

Financial Strategies

Homework

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Ongoing Costs of a Vehicle

Owning a vehicle involves regular costs beyond the initial purchase.

Cost	Frequency
Fuel	Weekly / ongoing
Registration	Annual
CTP insurance	Annual
Comprehensive insurance	Annual
Servicing / maintenance	Every 6–12 months
Tyres	Every few years
Loan repayments	Monthly (if applicable)
Tolls / parking	Ongoing

Calculating Ongoing Vehicle Costs

Fuel cost formula:

$$\text{Fuel cost} = \frac{\text{Distance (km)}}{100} \times \text{Consumption (L/100km)} \times \text{Price per litre}$$

Example

A car travels 12 000 km/year, uses 8.5 L/100km, and fuel costs \$2.10/L.

$$\frac{12\,000}{100} \times 8.5 \times 2.10 = \$2\,142 \text{ per year}$$

Total annual vehicle cost example:	Fuel	\$2 142
	Registration	\$380
	Insurance	\$1 750
	Servicing	\$600
	Total	\$4 872

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Reading Household Bills — Electricity

Electricity bills charge based on **kilowatt-hours (kWh)** consumed.

$$\text{Electricity cost} = \text{Usage (kWh)} \times \text{Rate (c/kWh)} + \text{Daily supply charge}$$

Example

Usage = 620 kWh over 90 days. Rate = 28.5c/kWh. Supply charge = 95c/day.

$$\text{Usage cost} = 620 \times \$0.285 = \$176.70$$

$$\text{Supply cost} = 90 \times \$0.95 = \$85.50$$

$$\text{Total} = \$176.70 + \$85.50 = \$262.20$$

Reading Household Bills — Water

Water bills typically include a **usage charge** (per kilolitre) and a fixed **service charge**.

$$\text{Water cost} = \text{Usage (kL)} \times \text{Rate (\$/kL)} + \text{Service charge}$$

Tiered pricing is common — higher usage attracts a higher rate.

Example

55 kL used. First 50 kL at \$2.35/kL, remainder at \$3.10/kL. Service charge = \$45.

$$\text{Tier 1} = 50 \times \$2.35 = \$117.50$$

$$\text{Tier 2} = 5 \times \$3.10 = \$15.50$$

$$\text{Total} = \$117.50 + \$15.50 + \$45.00 = \$178.00$$

Reading Household Bills — Gas

Gas is measured in **megajoules (MJ)** and billed similarly to electricity.

$$\text{Gas cost} = \text{Usage (MJ)} \times \text{Rate (c/MJ)} + \text{Daily supply charge}$$

Example

Usage = 980 MJ over 60 days. Rate = 3.2c/MJ. Supply charge = 80c/day.

$$\text{Usage cost} = 980 \times \$0.032 = \$31.36$$

$$\text{Supply cost} = 60 \times \$0.80 = \$48.00$$

$$\text{Total} = \$31.36 + \$48.00 = \$79.36$$

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Fixed vs Discretionary Spending

Fixed expenses are set amounts paid regularly — they do not change month to month.

- Rent / mortgage, loan repayments, insurance, subscriptions

Discretionary expenses are flexible — wants rather than needs.

- Dining out, entertainment, clothing, hobbies, holidays

A budget must satisfy:

$$\text{Total Income} \geq \text{Fixed Expenses} + \text{Discretionary Expenses} + \text{Savings}$$

Preparing a Personal Budget

Steps to build a budget:

1. Calculate total net (after-tax) income
2. List all fixed expenses
3. Estimate discretionary expenses
4. Allocate an amount to savings
5. Check: $\text{income} - \text{expenses} - \text{savings} \geq 0$

A common budgeting rule — the 50/30/20 rule:

- **50%** of income → needs (fixed expenses)
- **30%** of income → wants (discretionary)
- **20%** of income → savings

Budget Example

Weekly net income: \$1 050

Category	Item	Amount
Fixed	Rent	\$380
Fixed	Transport	\$60
Fixed	Phone / internet	\$35
Discretionary	Groceries	\$130
Discretionary	Dining / social	\$80
Discretionary	Entertainment	\$50
Savings	Emergency fund	\$100
Savings	General savings	\$115
Total		\$950
Remaining		\$100

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Strategies to Minimise Financial Problems

Four key strategies to maintain financial health:

1. **Budgeting** — plan how every dollar is spent before it is earned
2. **Saving** — consistently set aside a portion of income towards goals
3. **Emergency fund** — build a buffer of 3 to 6 months of living expenses
4. **Monitoring spending** — regularly review and adjust the budget

The Emergency Fund

An emergency fund covers unexpected expenses without going into debt.

Recommended amount: 3 to 6 months of living expenses.

Example

Monthly living expenses = \$2 400.

$$\text{Minimum fund} = 3 \times \$2\,400 = \$7\,200$$

$$\text{Ideal fund} = 6 \times \$2\,400 = \$14\,400$$

How to build it:

- Set a fixed weekly or fortnightly contribution
- Keep it in a separate, accessible savings account
- Only use it for genuine emergencies

Monitoring Spending

Regularly tracking spending helps identify where money is being wasted and whether the budget is being followed.

Useful strategies:

- Review bank statements weekly
- Use a budgeting app or spreadsheet
- Compare actual spending to budgeted amounts each month
- Adjust discretionary spending when overspending occurs

Signs of financial problems to watch for:

- Consistently spending more than income
- Relying on credit cards or BNPL to cover essentials
- No savings or emergency fund
- Missing bill or loan repayment deadlines

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Complete all questions from *Managing Money — Budgeting*.